

DRAMMEN BERSKOG, Norway

WMO: 014800

Lat: 59.7536N

Lon: 10.1236E

Elev: 8

StdP: 101.23

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-14.6	-20.0	0.6	-16.7	-17.3	0.8	-13.5	7.7	5.8	6.4	3.0	1.5	330	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	27.9	17.3	25.8	16.7	23.7	15.7	18.8	24.6	17.8	23.2	17.0	21.8	2.7	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.8	12.0	20.5	15.9	11.3	19.6	14.9	10.6	19.0	53.2	24.7	50.3	23.3	47.7	21.7	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.1	5.1	4.4	-18.5	30.2	3.3	2.3	-20.9	31.9	-22.8	33.2	-24.6	34.5	-27.0	36.2		
			-18.8	19.8	3.3	1.3	-21.1	20.7	-23.0	21.5	-24.9	22.2	-27.2	23.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-3.9	-2.9	1.1	6.5	11.5	15.7	18.3	16.4	12.6	6.6	1.5	-2.7	(6)
(7)		DBStd	8.66	5.26	5.04	4.15	3.21	3.73	2.58	2.73	2.13	2.73	3.46	4.03	5.84	(7)
(8)		HDD10.0	1979	432	361	277	114	24	0	0	0	9	116	254	393	(8)
(9)		HDD18.3	4268	691	594	535	356	213	88	36	67	171	364	503	651	(9)
(10)		CDD10.0	803	0	0	0	8	72	170	257	198	88	9	1	0	(10)
(11)		CDD18.3	50	0	0	0	0	2	7	34	7	0	0	0	0	(11)
(12)		CDH23.3	508	0	0	0	0	46	113	304	44	1	0	0	0	(12)
(13)		CDH26.7	109	0	0	0	0	7	26	75	2	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	1.9	1.9	1.9	2.1	2.1	2.2	2.0	1.8	1.9	1.7	1.7	1.8	(14)
(15)	Precipitation	PrecAvg	848	62	42	45	48	61	79	81	91	90	95	93	59	(15)
(16)		PrecMax	1232	142	106	128	124	120	166	149	183	238	235	373	107	(16)
(17)		PrecMin	647	16	11	2	10	10	13	10	25	33	24	16	16	(17)
(18)		PrecStd	115	32	24	31	28	32	42	32	38	46	50	70	28	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	9.1	14.4	20.1	27.5	29.7	30.9	26.7	22.3	17.1	13.3	10.7	(19)
(20)			MCWB	5.4	4.5	7.3	11.3	15.8	16.8	18.9	16.9	15.5	12.3	10.2	7.6	(20)
(21)		2%	DB	6.2	6.7	11.3	17.9	24.2	26.7	28.9	24.5	20.8	15.1	11.2	8.5	(21)
(22)			MCWB	3.7	3.5	6.0	10.2	14.7	15.8	18.2	16.6	14.4	10.9	8.7	6.0	(22)
(23)		5%	DB	4.5	5.1	8.9	15.4	21.7	23.9	26.9	22.9	19.3	13.7	9.2	6.8	(23)
(24)			MCWB	2.6	2.7	4.2	8.7	13.3	14.9	17.5	15.9	13.7	10.7	7.3	4.8	(24)
(25)		10%	DB	3.0	3.5	7.0	13.1	18.9	21.9	24.7	21.3	17.8	12.3	7.6	5.0	(25)
(26)			MCWB	1.3	2.0	3.5	7.4	12.3	13.6	16.9	15.2	13.1	9.5	6.0	3.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	5.6	8.2	12.4	17.6	18.4	20.6	18.8	17.1	13.7	11.2	8.1	(27)
(28)			MCDWB	7.9	7.7	12.8	18.2	23.8	26.2	27.1	23.0	19.8	16.0	12.7	10.3	(28)
(29)		2%	WB	4.2	4.0	6.3	10.6	16.1	17.1	19.5	17.9	15.7	12.3	9.1	6.3	(29)
(30)			MCDWB	5.7	5.8	10.2	16.4	21.6	23.2	26.0	21.8	19.1	14.1	10.7	8.1	(30)
(31)		5%	WB	2.9	3.0	5.1	9.4	14.6	16.0	18.5	17.0	14.7	11.2	7.6	4.9	(31)
(32)			MCDWB	4.3	4.7	8.0	14.3	20.0	21.7	24.5	20.9	17.8	13.2	9.0	6.5	(32)
(33)		10%	WB	1.4	2.0	3.8	8.1	13.1	15.0	17.6	16.2	13.9	9.9	6.2	3.1	(33)
(34)			MCDWB	2.7	3.4	6.6	12.3	17.3	19.9	23.1	19.9	16.7	11.7	7.4	4.6	(34)
(35)	Mean Daily Temperature Range			MDBR	5.4	6.1	8.8	10.2	10.3	10.0	9.3	8.9	7.0	5.4	5.6	(35)
(36)		5% DB	MCDBR	7.6	8.5	11.5	14.4	15.1	14.3	13.7	11.7	11.1	9.0	8.3	8.3	(36)
(37)			MCWBR	5.9	5.7	7.0	7.7	6.9	6.2	5.4	5.3	6.1	5.8	6.4	6.6	(37)
(38)		5% WB	MCDBR	7.2	7.3	10.8	12.8	12.6	11.7	11.8	9.4	9.5	7.6	7.6	7.9	(38)
(39)			MCWBR	5.9	5.2	6.9	7.1	6.1	5.8	5.2	5.0	5.7	5.6	6.2	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.258	0.298	0.324	0.368	0.366	0.355	0.376	0.369	0.337	0.323	0.280	0.232	(40)
(41)		taud		2.218	2.295	2.332	2.325	2.416	2.464	2.426	2.440	2.518	2.475	2.311	2.134	(41)
(42)		Ebn at Noon		541	712	809	832	863	878	847	825	798	686	514	420	(42)
(43)		Edh at Noon		41	66	87	105	103	100	102	93	73	57	38	29	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.31	0.93	2.44	3.87	4.90	5.59	4.95	3.91	2.58	1.17	0.41	0.20	(44)
(45)		RadStd		0.06	0.18	0.24	0.54	0.54	0.43	0.64	0.30	0.32	0.22	0.11	0.02	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (53 neighbors)	N/A	N/A	+1.56	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page